

Batch: C-3 / CCV 9 Roll No.: 16010123325

Experiment No. 1

Grade: AA / AB / BB / BC / CC / CD / DD

Title: Experiment 1: Study of Cloud Computing & Desktop-as-a-Service (DaaS)

Objective: To study the fundamentals of cloud computing and to understand the concept of Desktop-as-a-Service (DaaS) through exploration of free-access and demo-based platforms.

Prerequisites: Basic knowledge of computers, operating systems, and internet usage.

Expected Outcome of Experiment:

- Understanding of different computing models and cloud architecture.
- Knowledge of Desktop-as-a-Service and its applications.
- Ability to compare DaaS platforms and distinguish between DaaS and IaaS.

CO	Outcome
CO1	Comprehend the issues related to cloud computing and its application

Books/ Journals/ Websites referred:

<https://distrosea.com/>

<https://www.apporto.com>

<https://www.onworks.net>

<https://kasmweb.com>

Part A: Study of Cloud Computing Concepts (Theory)

1. Types of Computing

Traditional / Standalone Computing

Definition

Traditional computing refers to a system where all computation, storage, and applications run on a **single, independent machine** without relying on a network.

Examples

- Personal desktop computer
- Laptop running offline applications (MS Word, offline games)

Advantages

- Works without internet connectivity
- Simple setup and maintenance
- Better control over data locally
- Low initial complexity

Disadvantages

- Limited processing power and storage
- No resource sharing
- Data loss risk if the system fails
- Poor scalability

Client–Server Computing

Definition

Client–server computing is a model where **clients** request services or resources from a centralized **server** over a network.

Examples

- Email systems
- Web applications (browser as client, web server as server)
- Database systems in offices

Advantages

- Centralized data management
- Easier maintenance and updates
- Better security control
- Supports multiple users

Disadvantages

- Server failure affects all clients
- Higher setup and infrastructure cost
- Network dependency
- Server can become a bottleneck

Distributed Computing

Definition

Distributed computing involves multiple independent computers working together as a **single system** to solve a problem by sharing tasks and resources.

Examples

- Peer-to-peer networks
- Distributed databases
- Cluster computing systems

Advantages

- High performance through parallel processing
- Fault tolerance (system can continue if one node fails)
- Scalability by adding more nodes
- Efficient resource utilization

Disadvantages

- Complex system design
- Difficult debugging and maintenance
- Network latency issues
- Security challenges

Cloud Computing

Definition

Cloud computing provides **on-demand access** to computing resources (servers, storage, software) over the internet on a **pay-as-you-use** basis.

Examples

- Google Drive
- AWS EC2
- Microsoft Azure
- Online IDEs and SaaS platforms

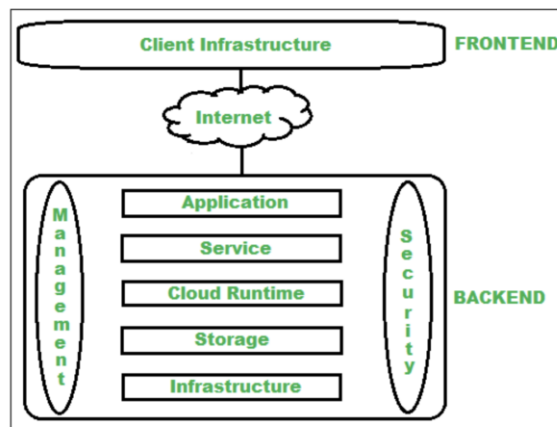
Advantages

- Highly scalable and flexible
- Reduced upfront cost
- Access from anywhere
- Automatic updates and maintenance

Disadvantages

- Internet dependency
- Data privacy and security concerns
- Vendor lock-in
- Ongoing operational cost

2. Cloud Computing Architecture



Following are the components of Cloud Computing Architecture:

1. **Client Infrastructure:** Client Infrastructure is a part of the frontend component. It contains the applications and user interfaces which are required to access the cloud platform. In other words, it provides a GUI(Graphical User Interface) to interact with the cloud.
2. **Application :** Application is a part of backend component that refers to a software or platform to which client accesses. Means it provides the service in backend as per the client requirement.
3. **Service:** Service in backend refers to the major three types of cloud based services like SaaS, PaaS and IaaS. Also manages which type of service the user accesses.
4. **Runtime Cloud:** Runtime cloud in backend provides the execution and Runtime platform/environment to the Virtual machine.
5. **Storage:** Storage in backend provides flexible and scalable storage service and management of stored data.
6. **Infrastructure:** Cloud Infrastructure in backend refers to the hardware and software components of cloud like it includes servers, storage, network devices, virtualization software etc.
7. **Management:** Management in backend refers to management of backend components like application, service, runtime cloud, storage, infrastructure, and other security mechanisms etc.
8. **Security:** Security in backend refers to implementation of different security mechanisms in the backend for secure cloud resources, systems, files, and infrastructure to end-users.
9. **Internet:** Internet connection acts as the medium or a bridge between frontend and backend and establishes the interaction and communication between frontend and backend.
10. **Networking:** Networking in backend services that provide networking infrastructure for application in the cloud, such as load balancing, DNS and virtual private networks.

Part B: Desktop-as-a-Service (DaaS)

1. Definition of DaaS

Desktop-as-a-Service (DaaS) is a **cloud computing service model** in which a **complete virtual desktop environment** (including operating system, applications, and user data) is hosted on **centralized cloud infrastructure** and delivered to end users over the internet.

Users can access their desktops remotely using a web browser or thin client, without requiring powerful local hardware.

2. Common Features of DaaS

Browser-Based Desktop Access: Users can access their virtual desktop directly through a **web browser**, eliminating the need for complex installations or local software dependencies.

Centralized Desktop Management: All desktops, applications, and updates are managed from a central server, making administration, monitoring, and maintenance easier for IT teams.

Device and Location Independence: DaaS desktops can be accessed from **any device** (laptop, tablet, thin client) and **any location**, as long as an internet connection is available.

Cloud-Based Data Storage: User data and files are stored securely in the **cloud**, reducing the risk of data loss due to device failure, and enabling easy backups and recovery.

Security and Access Control: DaaS platforms provide **authentication, encryption, role-based access control, and session isolation**, ensuring secure access to desktops and data.

Scalability: Resources such as CPU, memory, storage, and number of desktops can be **scaled up or down on demand**, making DaaS suitable for both small teams and large organizations.

Minimal Client-Side Requirements: End users only need a **basic device with internet access** since most processing and storage occur on the cloud servers.

Part C: Exploration of DaaS Platforms

Platform 1: OnWorks (Free Access)

Linux

Figure 1: Home Screen on OnWorks

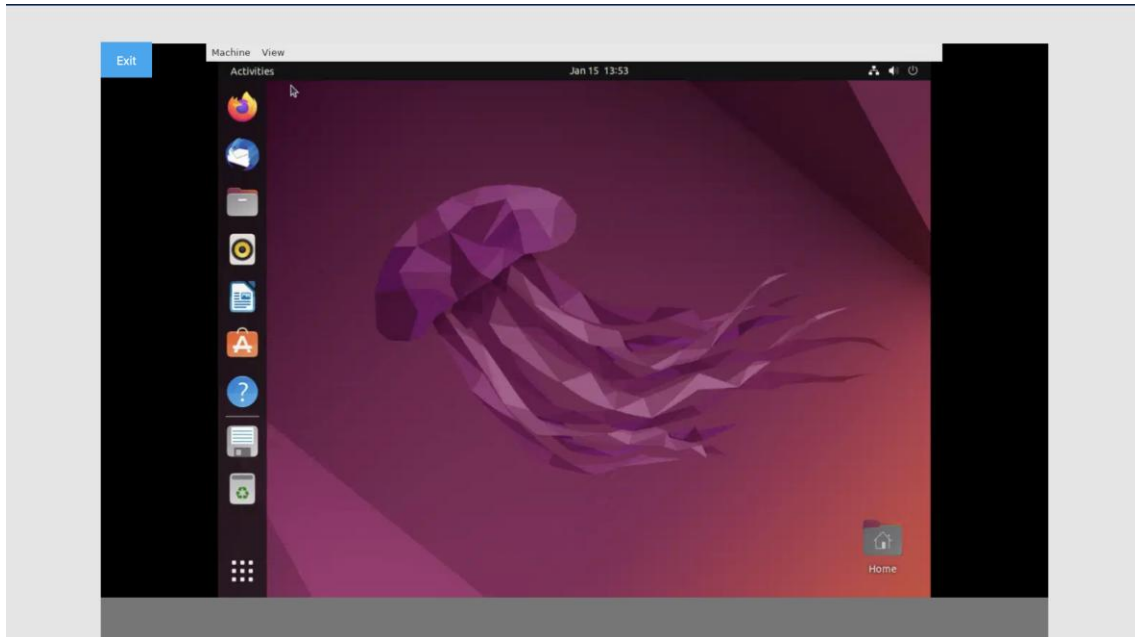


Figure 2: Application Menu

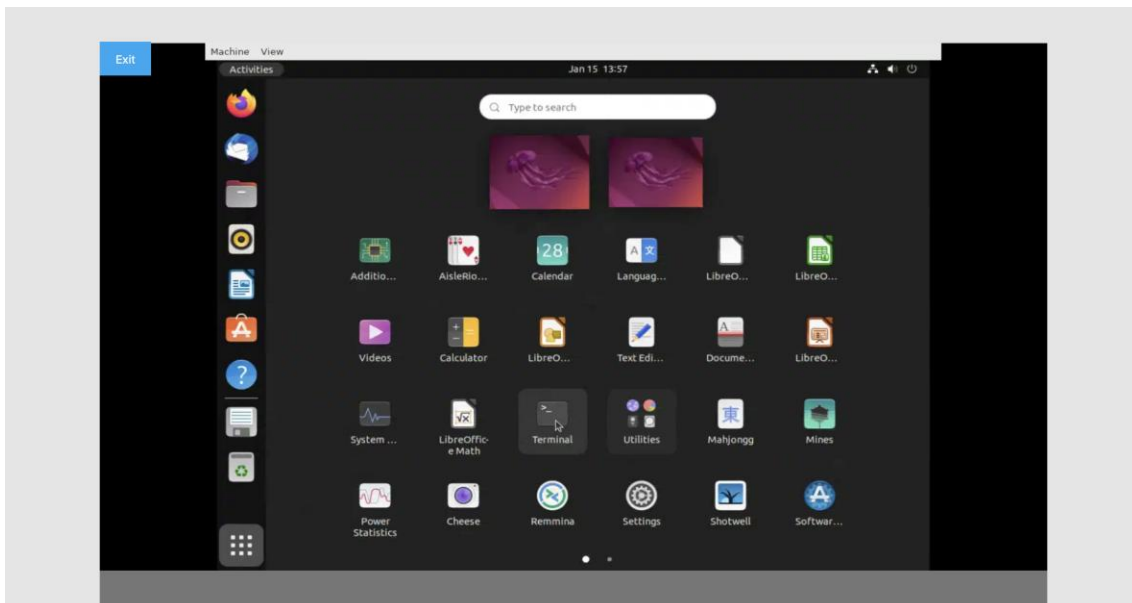


Figure 3: Using LibreOffice, Terminal & Firefox

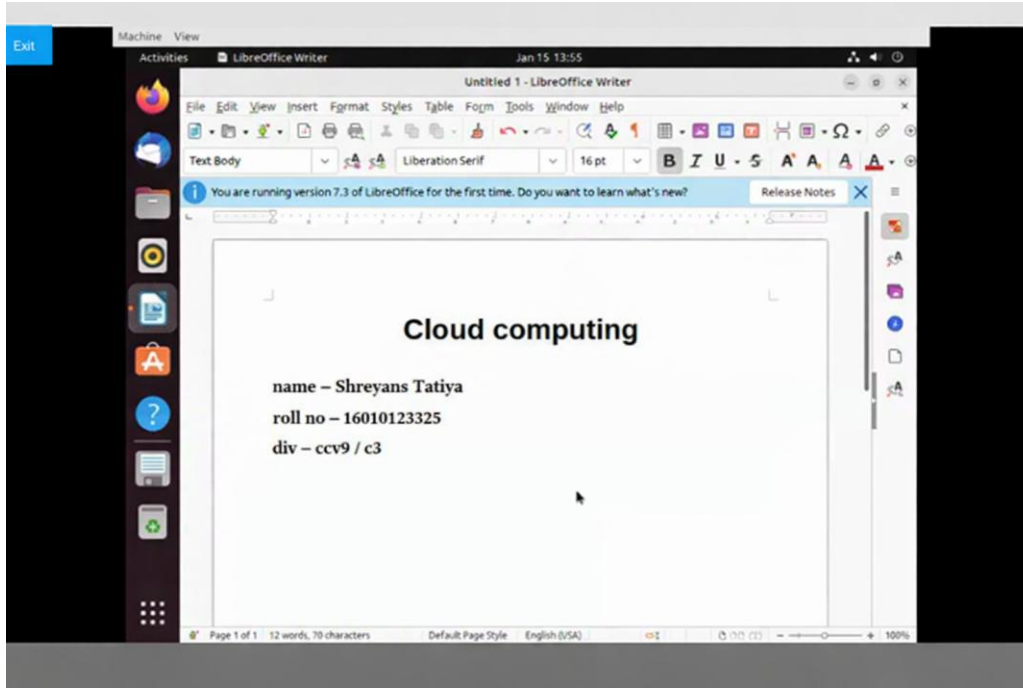


Figure 4: Terminal

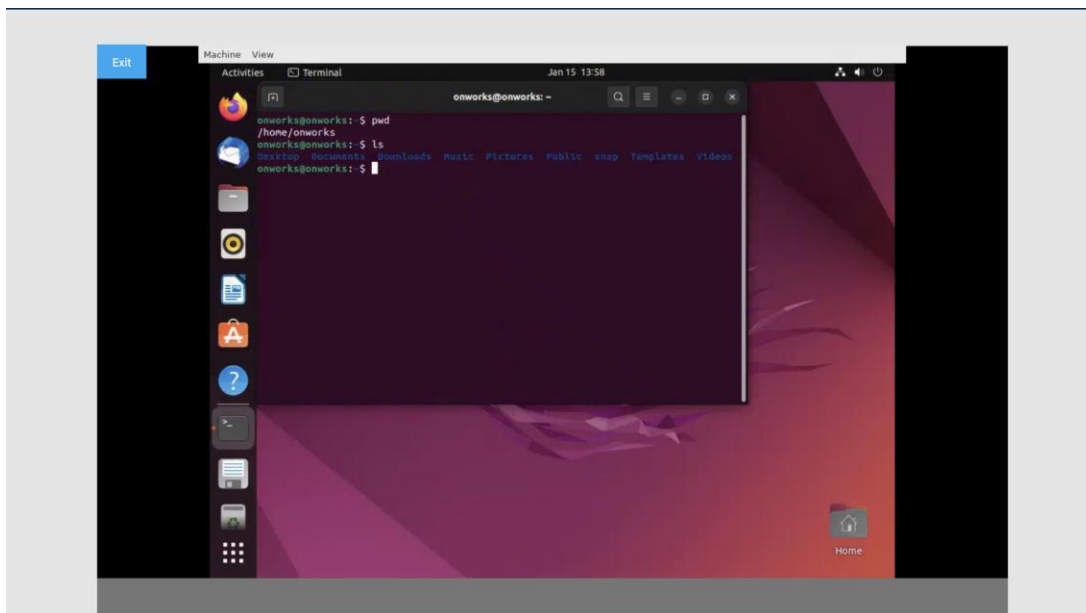


Figure 5: Firefox

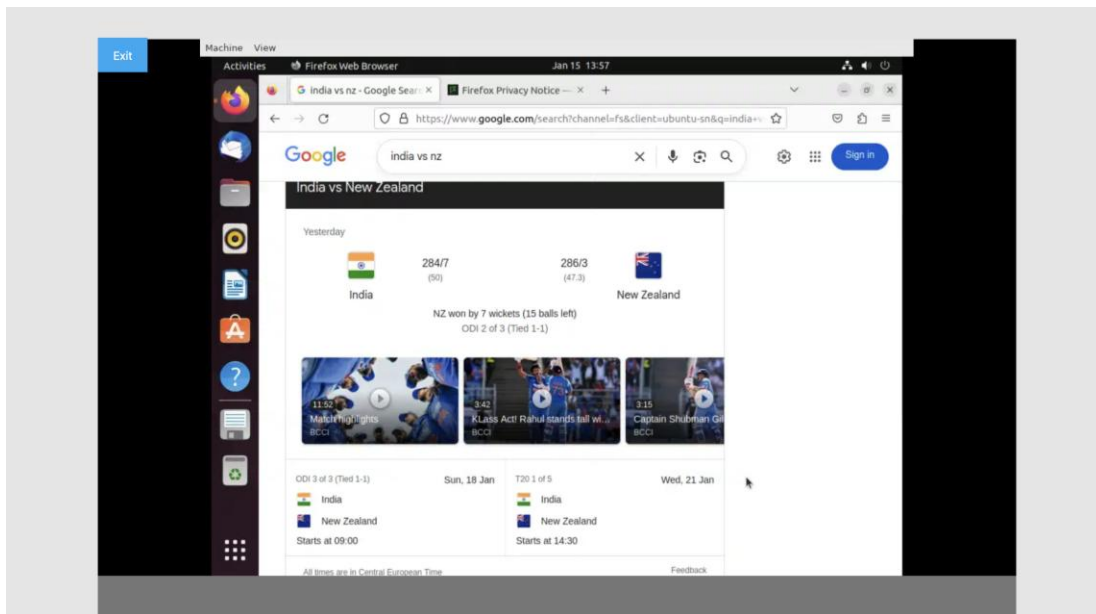
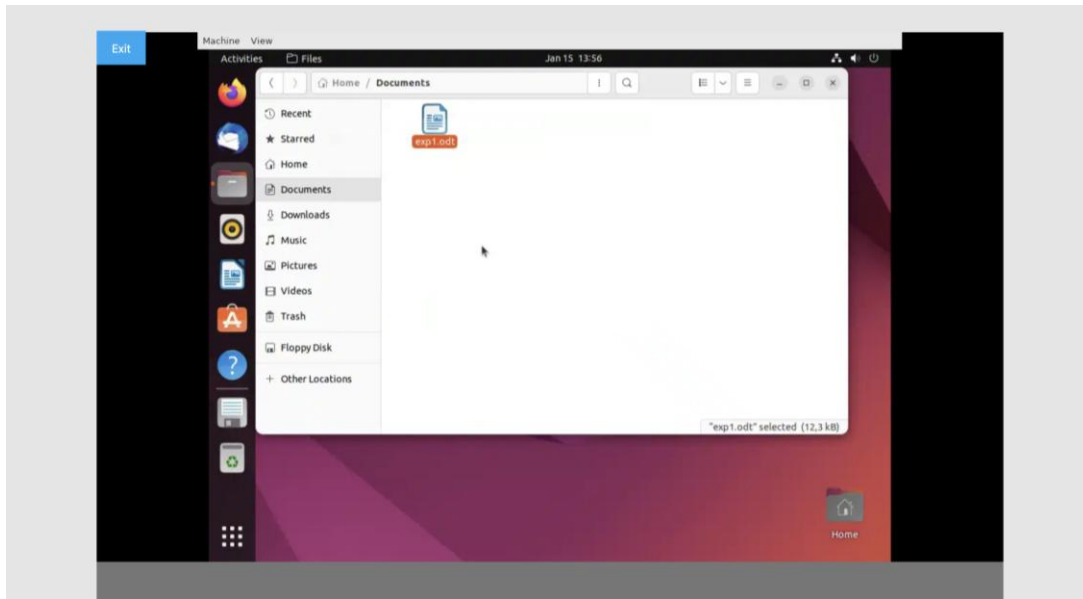


Figure 6: Files Application



Platform 2: Kasm Workspaces (Demo) Cloud Browser

Figure 7: Home Page

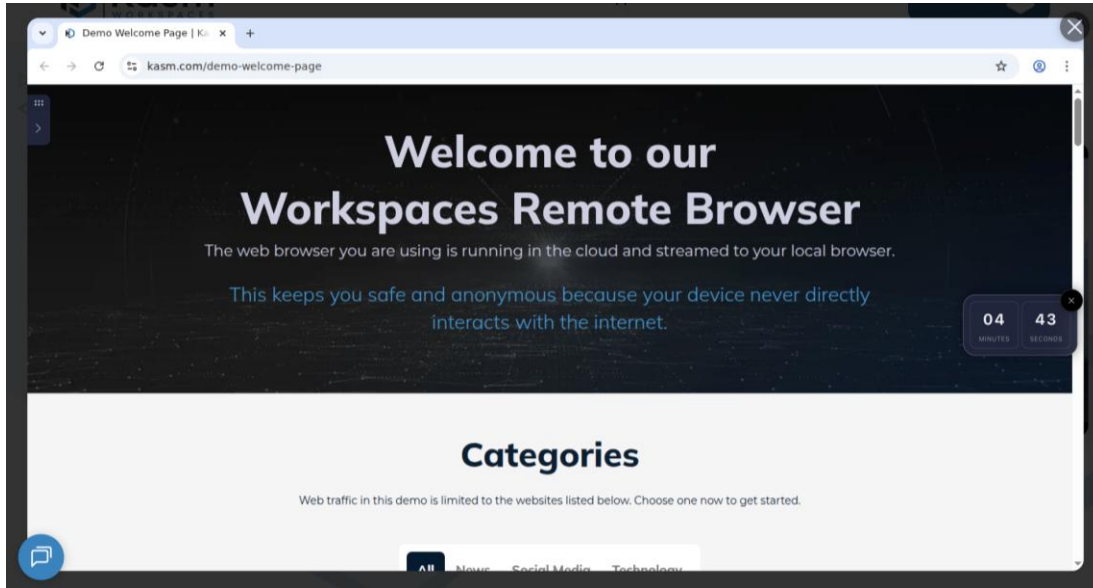
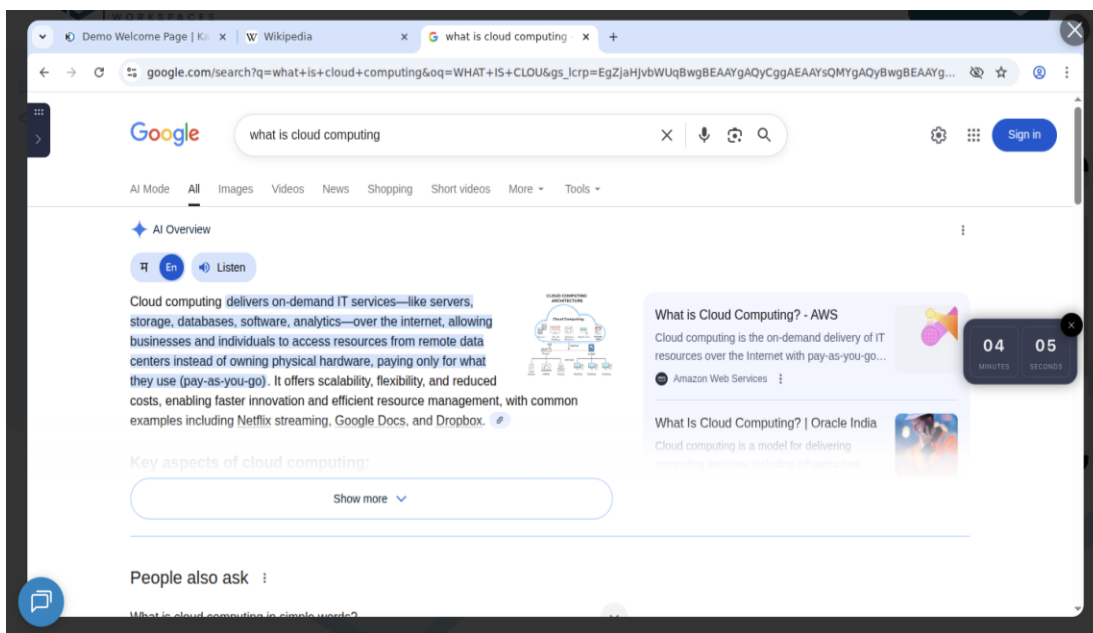
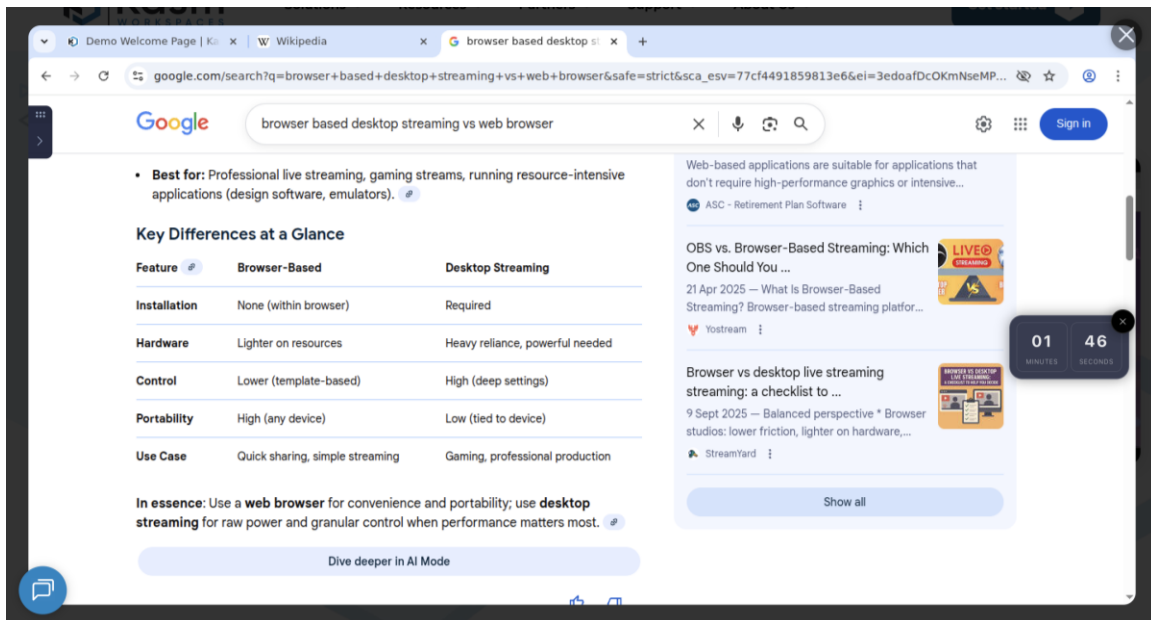


Figure 8: Google Search Engine





Demo Welcome Page | K J Somaiya School of Engineering | Wikipedia | browser based desktop streaming vs web browser

google.com/search?q=browser+based+desktop+streaming+vs+web+browser&safe=strict&scas_esv=77cf4491859813e6&ei=3edoafDcOKmNseMP...

Web-based applications are suitable for applications that don't require high-performance graphics or intensive...

• **Best for:** Professional live streaming, gaming streams, running resource-intensive applications (design software, emulators).

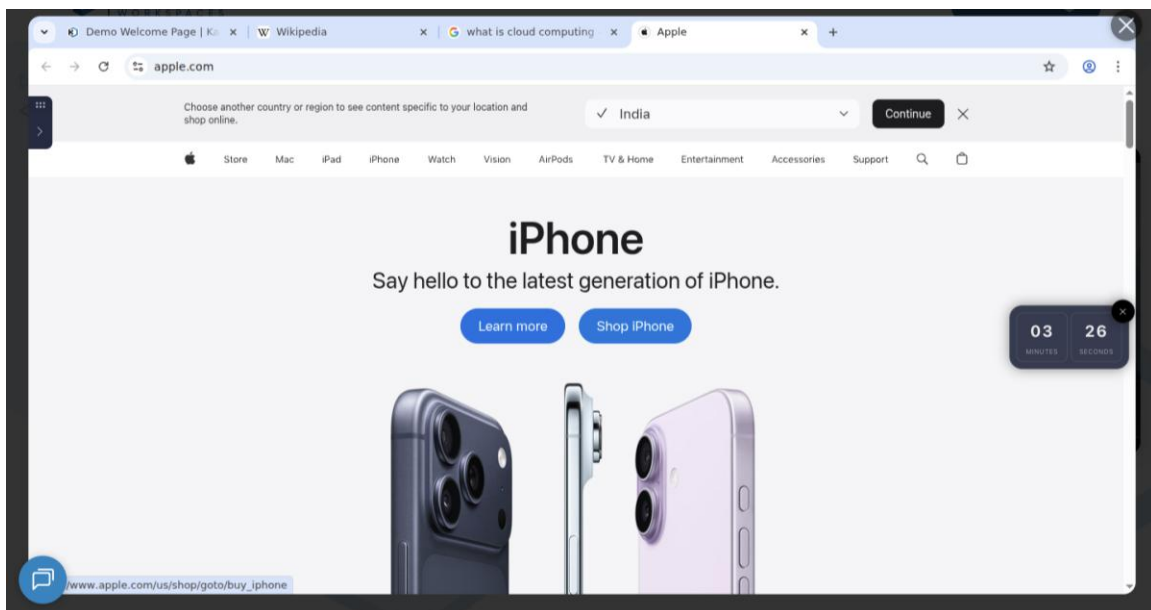
Key Differences at a Glance

Feature	Browser-Based	Desktop Streaming
Installation	None (within browser)	Required
Hardware	Lighter on resources	Heavy reliance, powerful needed
Control	Lower (template-based)	High (deep settings)
Portability	High (any device)	Low (tied to device)
Use Case	Quick sharing, simple streaming	Gaming, professional production

In essence: Use a **web browser** for convenience and portability; use **desktop streaming** for raw power and granular control when performance matters most.

Dive deeper in AI Mode

01 46 MINUTES SECONDS



Demo Welcome Page | K J Somaiya School of Engineering | Wikipedia | what is cloud computing | Apple

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iPhone

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03 26 MINUTES SECONDS

www.apple.com/us/shop/goto/buy_iphone

Platform 3: Distrosea

Fedora Linux

Figure 9: Home Screen

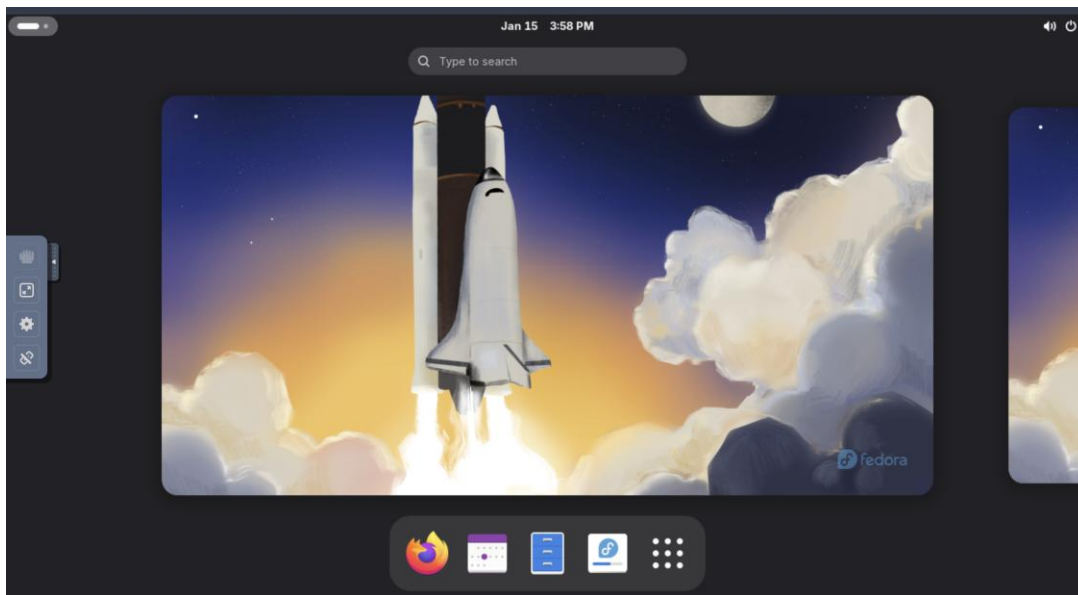


Figure 10: Application Library

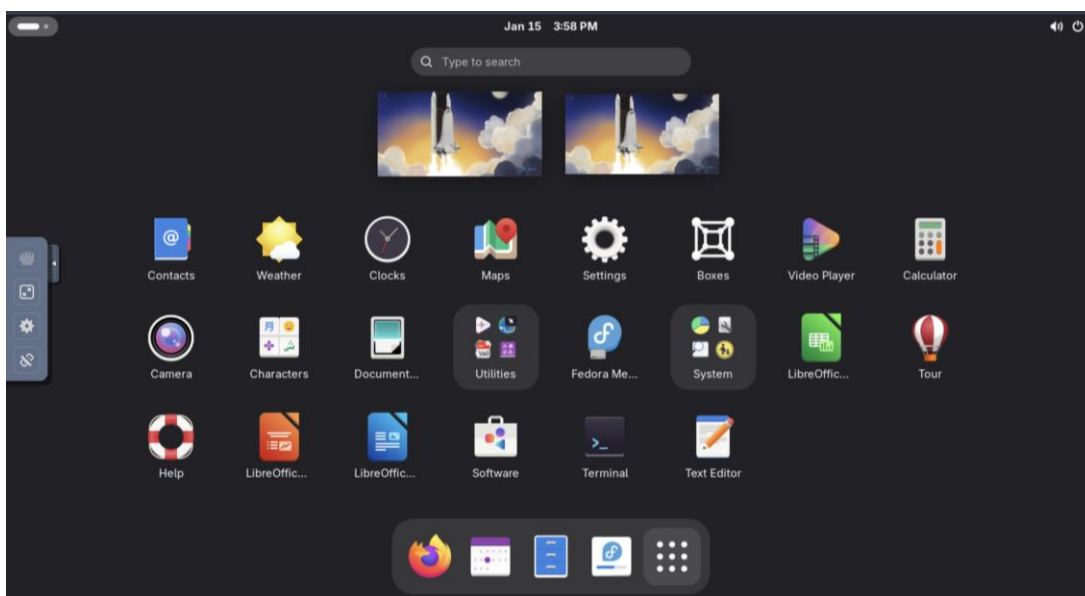


Figure 11: LibreOffice Writer

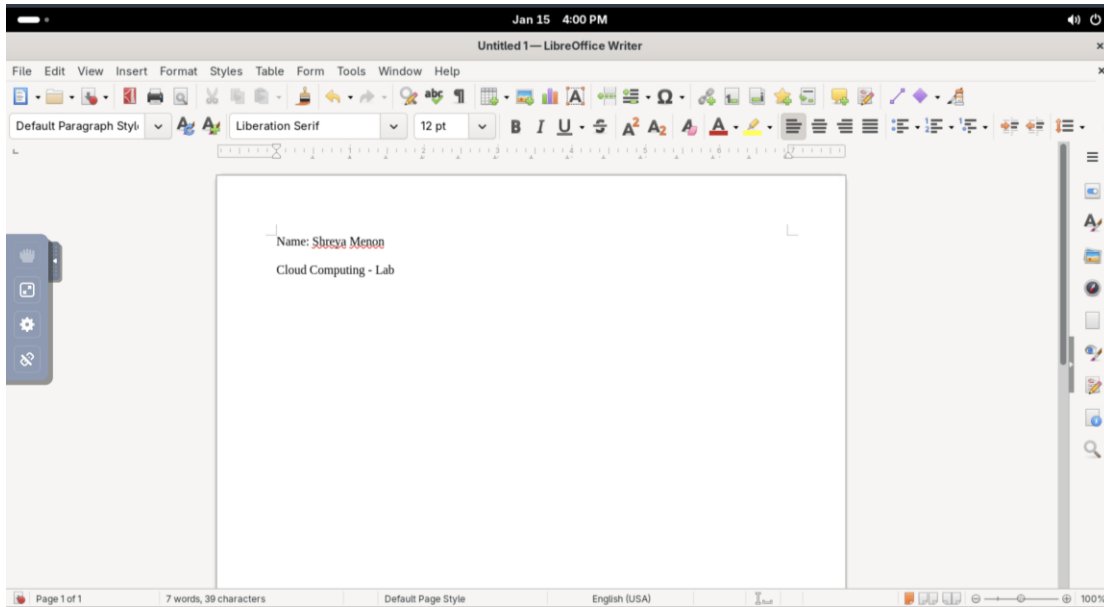
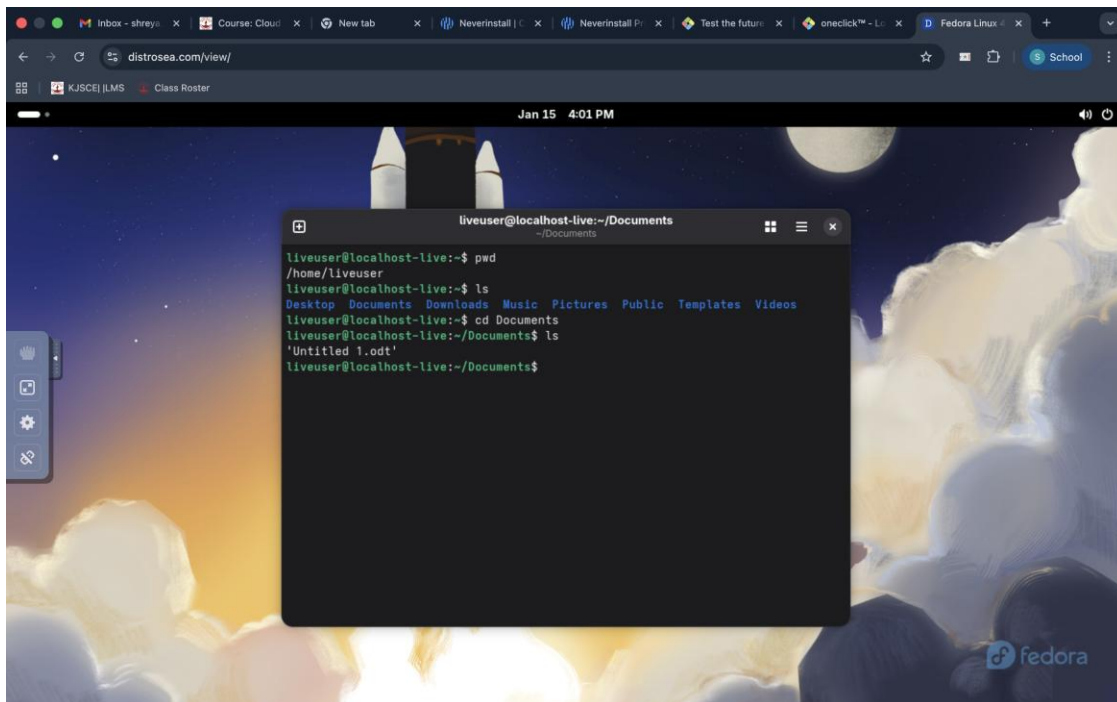


Figure 12: Terminal



Platform 3: Apporto

Windows

Figure 13: Home Screen



Figure 14: Bing Web Browser

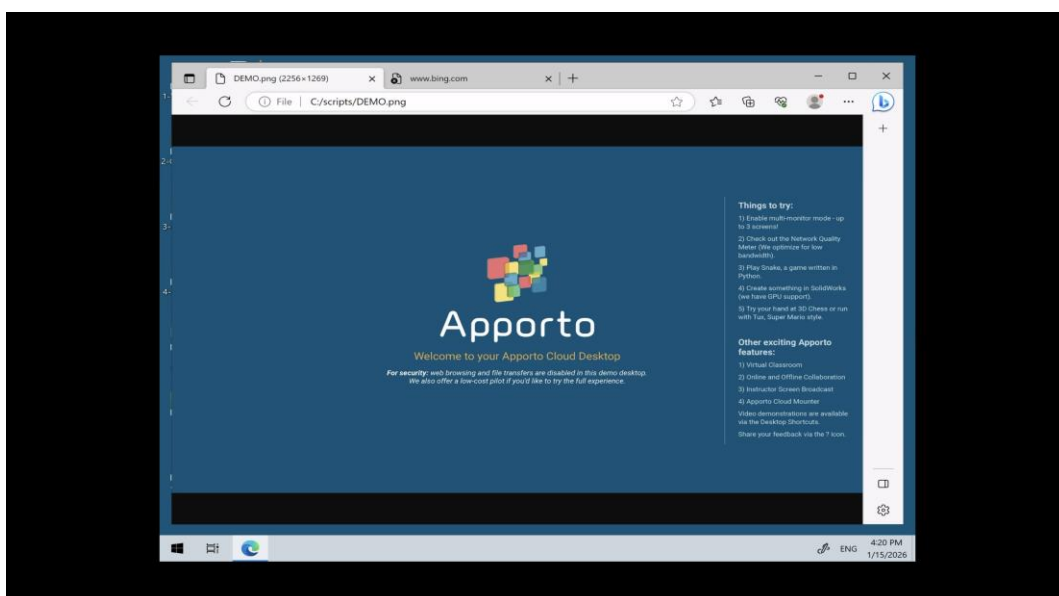
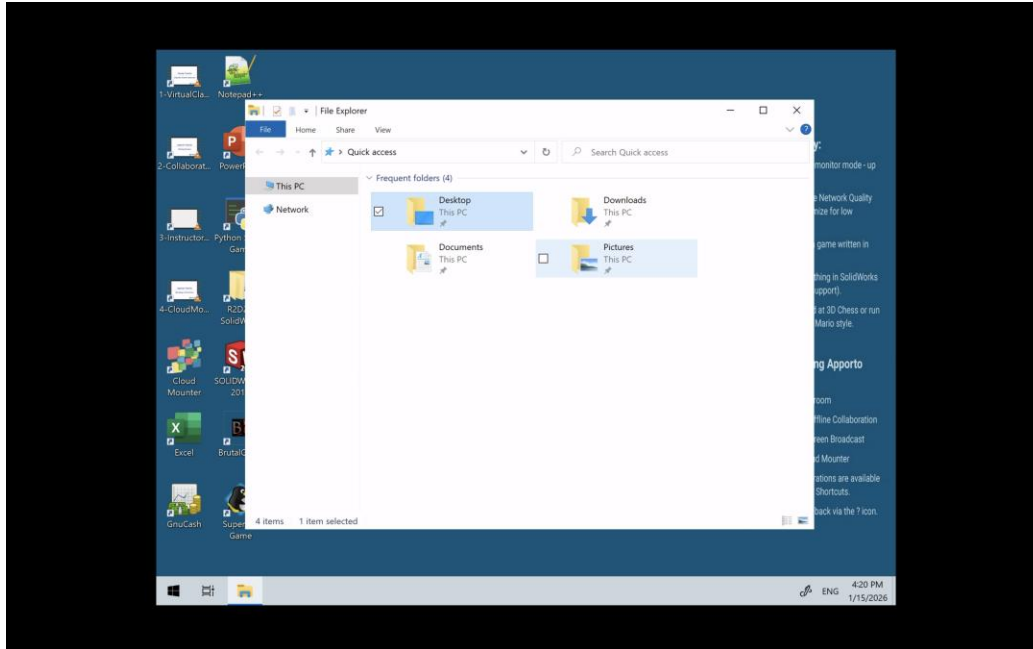


Figure 15: File Explorer



Part D: Comparative Analysis

Parameter	OnWorks	Kasm Workspaces	Apporto	DistroSea
Pricing model	Free (ad-supported)	Free (Community Edition) + Paid plans	Subscription-based (enterprise)	Free
Platform type	Free	Free / Demo / Paid	Paid + Demo (on request)	Free
Operating system support	Linux distros (Ubuntu, Fedora, Debian, etc.)	Linux & Windows desktops	Windows, Linux, macOS (managed desktops)	Linux distros only
Storage availability	Limited / session-based (not clearly persistent)	Ephemeral by default (session destroyed after use)	Configurable cloud storage (organization-managed)	Temporary / non-persistent
User interface	Web-based desktop via browser	Modern, polished web UI	Enterprise-grade browser-based UI	Simple noVNC browser interface
Ease of access	Very easy (launch directly in browser)	Easy (demo access; setup varies for enterprise)	Easy for users, requires org provisioning	Very easy (select distro and run)

Part E: Comparison between DaaS and IaaS

Parameter	Desktop as a Service (DaaS)	Infrastructure as a Service (IaaS)
Level of control	Limited control; users mainly manage applications and personal settings, while the provider manages OS, infrastructure, and maintenance	High control; users manage virtual machines, operating systems, storage, networking, and configurations
Target users	End users, students, remote workers, non-technical users, and organizations needing ready-to-use desktops	Developers, system administrators, IT teams, and organizations needing customizable infrastructure
Required expertise	Low; minimal technical knowledge required to use the virtual desktop	High; requires knowledge of system administration, networking, and cloud management
Examples	Apporto, Kasm Workspaces	Amazon EC2, Google Compute Engine

Result

Thus, the fundamentals of cloud computing were studied and the concept of Desktop-as-a-Service was understood using free-access and demo-based platforms.